

Lecture #20: Exit and Return Times

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Math 190: Mathematics of Sustainability, Emory University

April 22, 2026



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Goals of this lecture

By the end of this lecture, you should be able to

- Know what an expected time of return is,
- know how to calculate it, and
- know the relationships between stationary distributions and expected return time.



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① Limit Behaviors

② Return States

③ Exit Distributions



Catch-up from Last Lecture: Even-Odd Parity

Remark

We spent a lot of time talking about the even-odd parity of the Ehrenfest chain on Monday, but I forgot the most important part: We need to show that for this problem (because it is not aperiodic), that $\lim_{n \rightarrow \infty} p^n(x, y) \not\rightarrow \pi(y)$.

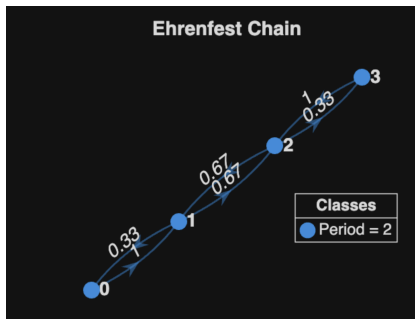


Figure 1: Ehrenfest Chain

Catch-up from Last Lecture: Doubly stochastic.

Question

If the matrix is doubly stochastic, then the uniform distribution $\pi(x)$ is uniform. Does it work the other way? I.E. if the stationary distribution is uniform, does that mean that the transition matrix is doubly stochastic.

Solution

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Notation Refresher

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 - we also denote $N(y)$ be the number of visits to y at times $n \geq 1$.



Question

What is the probability we go to y k times after starting at x ?

Proof.



Computing the expected value

First we want to show that we can compute the expected value of X , which is $\mathbb{E}X$.

Lemma

For any nonnegative integer valued random variable X , the expected value of X can be computed by

$$\mathbb{E}X = \sum_{k=1}^{\infty} \mathbb{P}(X \geq k).$$



Computing the expected value (cont.)

Proof.

We define $\mathbf{1}_{\{X \geq k\}}$ be a random variable that is 1 if $X \geq k$ and 0 otherwise.

From this example, we can see that

Now we take the expected value of both sides. But we notice that $\mathbb{E}\mathbf{1}_{\{X \geq k\}} = \mathbb{P}(X \geq k)$, so



Lemma

$$\mathbb{E}_x N(y) = \frac{\rho_{xy}}{1 - \rho_{yy}}$$

Proof.

Now the probability of returning at least k times, $\{N(y) \geq k\}$, is the same as the event that the k^{th} return occurs, i.e., $\{T_y^k < \infty\}$, so using Eqn 1, we have

The last equality happens because of the convergence of a geometric series, which we will show now. □



Convergence of a geometric series

Consider the geometric series:

Multiply both sides by r :

Now subtract the second equation from the first:

Most terms cancel:



Convergence of a geometric series (cont.)

Factor the left-hand side:

Solve for S :

If $|r| < 1$, taking the limit as $n \rightarrow \infty$:

For our problem, we have $r = \rho_{yy}$, which has to be smaller than 1, so



Computing the expected number of returns to y in a different way

Lemma

$$\mathbb{E}_x N(y) = \sum_{n=1}^{\infty} p^n(x, y).$$

Proof.

We start by reintroducing this notation $\mathbf{1}_{\{X_n=y\}}$ meaning that the variable is 1 only if $X_n = y$ and zero every other time. This means that

Taking the expected values gives



Recurrent States are likely to return infinitely often.

Theorem

y is recurrent if and only if

$$\sum_{n=1}^{\infty} p^n(y, y) = \mathbb{E}_y N(y) = \infty.$$

Proof.

The first equality comes from what we just showed. The only difference is we start at y and not x . The second comes from the geometric series equation and $\rho_{yy} = 1$ if it is recurrent, so



Relating the stationary distribution and expected return time.

Theorem

Suppose that a Markov Chain is irreducible and all the states are recurrent. If $N_n(y)$ be the number of visits to y up to time n , then

$$\frac{N_n(y)}{n} \rightarrow \frac{1}{\mathbb{E}_y T_y}$$

Theorem

If the Markov chain is irreducible and there is a stationary distribution, then

$$\pi(y) = \frac{1}{\mathbb{E}_y T_y}$$



Relating the stationary distribution and expected return time. (cont.)

Proof.

Let X_0 has distribution π . From above, we see that

$$\frac{N_n(y)}{n} \rightarrow \frac{1}{\mathbb{E}_y T_y}.$$

Taking expected value and using the fact that $N_n(y) \leq n$, it can be shown that this implies

but since π is a stationary distribution



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Example of Exit Times

Example

At a local two year college, 60% of fresh- men become sophomores, 25% remain freshmen, and 15% drop out. 70% of sophomores graduate and transfer to a four year college, 20% remain sopho- mores and 10% drop out. What fraction of new students eventually graduate?

Solution

We use a Markov chain with state space $1 = \text{freshman}$, $2 = \text{sophomore}$, $G = \text{graduate}$, $D = \text{dropout}$. The transition probability is

Exit Times Continued

	1	2	G	D
1	0.25	0.6	0	0.15
2	0	0.2	0.7	0.1
G	0	0	1	0
D	0	0	0	1

Let $h(x)$ be the probability that a student currently in state x eventually graduates. By considering what happens on one step.

To solve we note that the second equation implies $h(2) = 7/8$ and then the first that

